

# Plug-and-Play Image Restoration with Deep Denoiser Priors

## IRCNN vs DRUNet (and our novel IRCNN+)

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# Context & Goal of the Project

## The Classical Inverse Problem

$$y = Hx + n \quad \Rightarrow \quad \hat{x} = \arg \min_x \frac{1}{2} \|y - Hx\|_2^2 + \lambda \phi(x)$$

### Model-based methods

- + Flexible (change  $H$ )
- Limited explicit priors

### End-to-end learning

- + Very strong performance
- Strictly task-specific

## The Plug-and-Play (PnP) Paradigm

Replace the explicit prior  $\phi(x)$  with a **deep learned denoiser** (trained once, reused across all tasks).

### Our Project contribution:

- Implemented PnP-HQS from scratch for 4 tasks: *Denoising*, *Deblurring*, *SISR*, *Inpainting*.
- Implemented, trained & evaluated 3 models from scratch: *IRCNN*, *DRUNet*, *IRCNN+*.

HQS variable splitting with an auxiliary variable  $z$ :

$$\mathcal{L}_\mu(x, z) = \frac{1}{2} \|y - Hx\|_2^2 + \lambda\phi(z) + \frac{\mu}{2} \|z - x\|_2^2$$

Alternating scheme:

$$\begin{aligned} x^{k+1} &\leftarrow (H^\top H + \mu_k I)^{-1} (H^\top y + \mu_k z^k) \quad (\text{data fidelity step}) \\ z^{k+1} &\leftarrow D_{\sigma_k}(x^{k+1}), \quad \sigma_k \propto \sqrt{\lambda/\mu_k} \quad (\text{prior step}) \end{aligned}$$

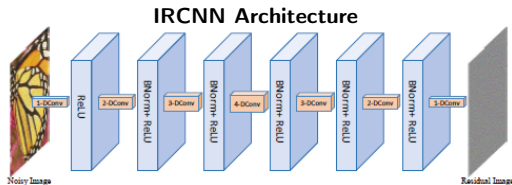
**Key Takeaways:**

- **Modularity:** Only  $H$  changes per task (blur, downsample, mask). The denoiser is universal.
- **Noise Schedule:**  $\sigma_k$  decreases logarithmically ( $\mu_k$  increases) to force  $x \approx z$  at convergence.

# The 3 Deep Denoisers

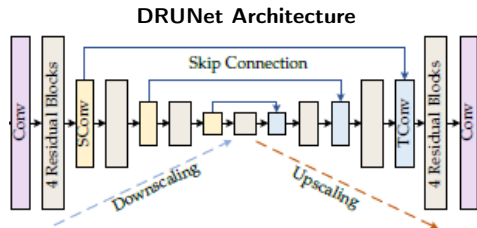
## IRCNN (bank of experts)

- 25 CNNs (7-layer dilated)
- $\sigma \in \{2, 4, \dots, 50\}$
- Inference: pick closest expert



## DRUNet ( $\sigma$ -map conditioning)

- Single model (residual U-Net)
- input: image + **sigma-map**
- handles a continuous noise range



## IRCNN+ (our idea)

- same backbone as IRCNN
- but **single model** + sigma-map
- goal: avoid expert selection

*Non-blind setting:  $H$  and  $\sigma$  are assumed known.*

# Training Strategy & Learning Targets

## Training Modalities

- **Implementation:** Models built and trained entirely from scratch.
- **Dataset:** BSDS300 (400 natural images).
- **Data Augmentation:** Horizontal/vertical flips and 90° rotations applied to prevent overfitting and improve generalization.

## Learning Objectives: DRUNet vs. IRCNN

- **DRUNet (Direct Mapping):** The network learns to predict and output the **clean, denoised image** directly.
- **IRCNN & IRCNN+ (Residual Learning):** The network learns to predict the **noise residual**. The denoised image is then obtained by subtracting this predicted noise from the original noisy input.

# Quantitative Results Analysis

PSNR (dB)

| Method   | Den.       | Deb.       | SISR       | Inp.       |
|----------|------------|------------|------------|------------|
| Degraded | 20.53±0.26 | 22.70±2.29 | 25.25±2.59 | 29.03±2.90 |
| IRCNN    | 29.64±1.36 | 27.39±1.00 | 27.80±2.72 | 29.80±1.42 |
| IRCNN+   | 29.21±1.45 | 28.06±1.98 | 29.32±2.80 | 26.59±2.18 |
| DRUNet   | 30.61±2.08 | 29.68±2.40 | 29.53±2.91 | 31.25±2.91 |

SSIM

| Method   | Den.      | Deb.      | SISR      | Inp.      |
|----------|-----------|-----------|-----------|-----------|
| Degraded | 0.40±0.11 | 0.47±0.10 | 0.67±0.12 | 0.89±0.01 |
| IRCNN    | 0.82±0.04 | 0.77±0.04 | 0.74±0.12 | 0.88±0.03 |
| IRCNN+   | 0.83±0.03 | 0.79±0.04 | 0.84±0.07 | 0.84±0.02 |
| DRUNet   | 0.86±0.05 | 0.84±0.06 | 0.85±0.08 | 0.92±0.01 |

## Key Insights

- **DRUNet Dominance:** Best performance across all tasks as expected, especially on **Inpainting**.
- **Denoising:** IRCNN outperforms IRCNN+ in PSNR but not in SSIM, proving that IRCNN+ provides good images for human perception even if it's not perfect mathematically.
- **Deblurring & SISR:** Surprisingly, IRCNN+ surpasses IRCNN, showing superior structural recovery.
- **Inpainting Contrast:** IRCNN slightly improves initialization, while IRCNN+ is in total failure due to over-smoothing.

## Qualitative results — Denoising ( $\sigma = 20$ )



Noisy



IRCNN

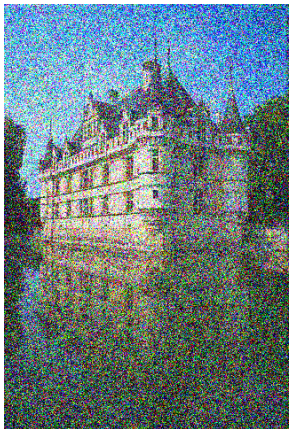


IRCNN+

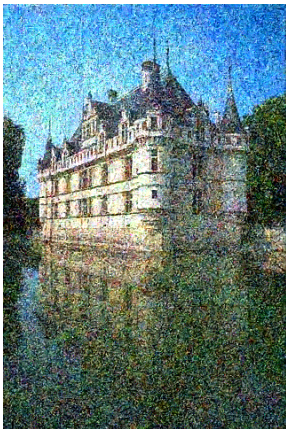


DRUNet

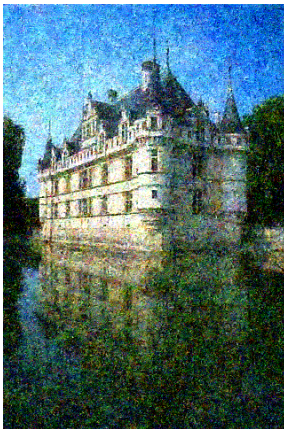
# Qualitative results — Denoising ( $\sigma = 100$ , CASTEL, out-of-range)



Noisy



IRCNN



IRCNN+



DRUNet



# Qualitative results — Deblurring (Levin kernel, $\sigma = 5$ , PLANE)



Blurred



IRCNN



IRCNN+



DRUNET

# Qualitative results — Inpainting (PLANE)



Masked image



IRCNN



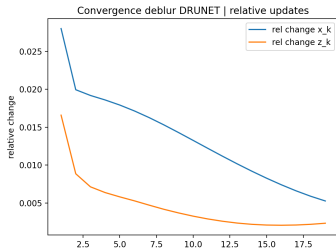
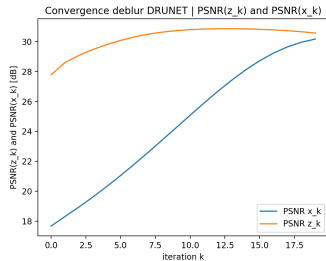
IRCNN+



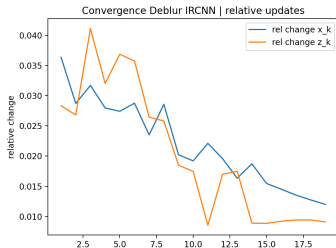
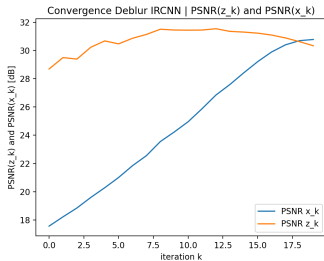
DRUNET

# Convergence

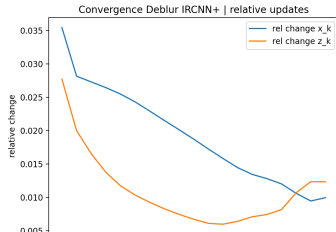
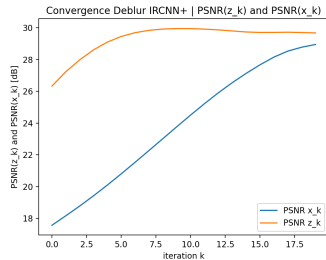
## DRUNet (deblur)



## IRCNN (deblur)



## IRCNN+ (deblur)



## The Power of PnP-HQS

A **generic and modular framework**: a single solver where only the degradation operator  $H$  changes across tasks.

### Model Performances

- **DRUNet**: Best overall. Superior stability thanks to U-Net architecture and continuous  $\sigma$ -map.
- **IRCNN**: Competitive but limited by its *discrete experts* which can introduce optimization shocks.

### Our Contribution: IRCNN+

- + Removes expert switching (more efficient training/inference).
- + Solid results on Deblurring/SISR.
  - Sensitive to strong data constraints (Inpainting failure).



K. Zhang, W. Zuo, S. Gu, L. Zhang.

*Learning Deep CNN Denoiser Prior for Image Restoration.*  
CVPR, 2017.



K. Zhang et al.

*Plug-and-Play Image Restoration with Deep Denoiser Prior.*  
TPAMI, 2021. (DRUNet / DPIR)